

Haoyuan Zheng

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EDUCATION

University of Michigan, Ann Arbor Sep. 2026 – present
B.S.E. in Computer Engineering

Shanghai Jiao Tong University (SJTU), Shanghai Sep. 2024 – Jun. 2028 (expected)
B.S.E. in Electrical and Computer Engineering | GPA: 3.68/4.0

Core courses: Honors Mathematics IV(A+), Intro to Circuits(A+), Physics II(A), Signal & System(A-), Data Structure and Algorithms

Awards: SJTU Undergraduate Class C Excellence Scholarship 2025; Cheng Family Scholarship 2026

RESEARCH EXPERIENCES

Undergraduate Researcher, School of Electrical Information and Electrical Engineering, SJTU

Design and Implementation of a Distributed Multi-Agent System for Novel Generation Nov. 2025 – Apr. 2026
Advisor: Jiong Lou, Research Assistant Professor, Department of Computer Science and Engineering, SJTU

- Designed C2-MANS, a neuro-symbolic multi-agent system integrating Large Language Models with Pydantic state models to produce logically consistent long-form narratives.
- Architected an iterative pipeline using CrewAI that employs four specialized agents to enforce cross-chapter consistency through scene-level constraints and post-generation state synchronization.
- Developed a modular Python codebase featuring persistent storage and structured state management layers such as RelationshipGraph and TimeLedger to support the autonomous generation of 10-chapter novellas.

Undergraduate Research Project, Institute of Robotics and Machine Vision (IRMV), SJTU

3D Vision and Visual SLAM Research Project Jul. 2026 – Aug. 2026
Advisor: Hesheng Wang, Tang Junyuan Chair Professor and Dean of Global College, Shanghai Jiao Tong University

- Built an object-oriented visual SLAM system in Python, implementing SIFT/ORB feature extraction, geometric matching, relative pose estimation, PnP tracking, triangulation, and sparse bundle adjustment.
- Conducted controlled experiments on frame sampling, keyframe selection, feature density, and match filtering, comparing reprojection error, point-cloud density, and runtime across different configurations.
- Extended the pipeline toward dense 3DGS-SLAM by integrating Depth Anything V2 and ADD-SLAM, and diagnosed failures caused by limited viewpoint coverage, depth-scale inconsistency, and camera-pose drift.

SELECTED PROJECTS

Hypertension Risk Analysis and Logistic Regression--STAT1000J Data Science Project Jul. 2025

- Built a Python data-analysis pipeline for 1,985 health records, including data cleaning, categorical encoding, feature scaling, visualization, and permutation-based hypothesis testing.
- Trained and evaluated a logistic-regression model with stratified splitting, achieving 75.31% test accuracy; analyzed feature coefficients, prediction errors, and confidence levels.

SKILLS

Programming: Python, C/C++, MATLAB

Computer Vision & Robotics: Multi-Agent Systems, 3D Reconstruction, Multi-View Geometry, Visual SLAM.

Tools & Libraries: PyTorch, Pandas, Git, Linux, LaTeX